PSAI_139

SCALE Codec Comparator

- **Team Name:** gmajor
- **Payment Address:** 0xC3094f0ddce699a1Ad9Ef2621DF68Cd297a4c44F(Dai)
- **Level:** 2

Project Overview :page_facing_up:

Overview

This Proposal is based on https://github.com/w3f/Grants-Program/blob/master/rfps/scale-codec-comparator.md. To date, there are more than 9 published implementations of the SCALE Codec. Since each is implemented by a different team & the reference implementation still introduces small fixes, it would be beneficial to compile a table of feature-completeness. This would provide (some) assurance that the implementation in a given language is safe & sound to use.

I have already developed https://github.com/gmajor-encrypt/php-scale-codec && https://github.com/gmajor-encrypt/php-substrate-api in the previous grant.

Project Details

1. Providing an FFI to Rust's reference implementation
2. Tested with rust's FFI along with the scale lib https://docs.substrate.io/reference/scale-codec/
3. Passing the test will release the scale code badge of the GitHub action
4. Add Github action ci automatically test if these libs are updated

Ecosystem Fit

Help developers choose suitable the scale code package in the ecosystem to avoid unknown errors

Team :busts_in_silhouette:

Team members

- gmajor

Contact

- **Contact Name:** gmajor
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- **Website:** https://github.com/gmajor-encrypt

Legal Structure

individual

Team's experience

I have many years of php development experience and nearly five years of blockchain development experience, familiar with PHP, GOLANG, PYTHON, Nodejs, Rust

Team Code Repos

https://github.com/gmajor-encrypt/php-scale-codec

https://github.com/gmajor-encrypt/php-substrate-api

Development Status :open_book:

Not yet

Development Roadmap :nut_and_bolt:

Overview

- **Total Estimated Duration:** 3.5 months
- **Total Costs:** 12500 DAI

Milestone 1

- **Estimated duration:** 2 month
- **FTE:** 1
- **Costs:** 5000 USD

Number	Deliverable	Specification
0a.	License	MIT
0b.	Documentation	Simple documentation on how to use and how to test
1.	enable Unit test	Providing a FFI make test for scale.go, php-scale-codec,scale.rb,py-scale-codec,polkadot-js
2.	github action	Auto test when comparator commit code

Milestone 2

- **Estimated duration:** 2 month
- **FTE:** 1
- **Costs:** 5000 USD

Number	Deliverable	Specification
0a.	License	MIT
0b.	Documentation	Simple documentation on how to use and how to test
1.	enable Unit test	Providing a FFI make test for as-scale-codec,cScale,scale-codec-cpp,scale-codec-js-library,go-substrate-rpc-client,scale-ts
2.	github action	Auto test when comparator commit code

Milestone 3

- **Estimated Duration:** 2 week
- **FTE:** 1
- **Costs:** 2,500 USD

Number	Deliverable	Specification
0a.	License	MIT
0b.	Documentation	Simple documentation on how to use and how to test
1.	GitHub badge	Create a GitHub badge for SCALE feature complete libs
2.	GitHub Ci ensure badge	Libs are tested with github actions if they are released

Future Plans

I have been maintaining php scale code and php api lib for 2 years, and this application will continue to be maintained